Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/12/2009
PI Name:	Kucik, Dennis F. M.D., Ph.D.		
Project Title:	Effect of high energy particle irradiation on adhesiveness of vascular endothelium and its consequences for atherosclerosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
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Zip Code:	35205-4831	**Congressional District:**	7
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: changed to Cardiovascular Alterations Team (from Radiation Effects Team) effective 5/1/2008, per K. Major/NSBRI (6/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kabarowski, Janusz (University of Alabama at Birmingham)		
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Task Description:	Inflammation plays a major role in radiation damage to tissues. Chronic, low-level vascular inflammation is a risk factor for atherosclerosis. Consistent with this, exposure to radiation from a variety of sources is associated with increased risk of heart disease and stroke. At least part of the inflammatory response to radiation is a change in the adhesiveness of the endothelial cells that line the blood vessels, triggering inappropriate accumulation of white blood cells. This leads to later, damaging effects of inflammation. Increased adhesion in response to radiation differs from physiological activation of adhesion in many ways, though, so predicting the consequences of radiation-induced endothelial cell adhesiveness is uncertain at our present state of knowledge. Although some studies have been done on the effects of gamma irradiation (the kind used for cancer treatment) on vascular endothelium, the response of endothelium to the high energy radiation likely to be encountered in prolonged space flight has not been determined. A systematic characterization will allow prediction of pathological consequences, including atherosclerosis, reducing the uncertainty in risk prediction and pointing the way to possible countermeasures. Our hypothesis is that high energy heavy ions alter the adhesive properties of vascular endothelium, and resultant vascular inflammation accelerates atherosclerosis. We are testing this hypothesis using both isolated cells and whole mice to determine the role of key adhesion molecules in recruiting white blood cells to sites of inflammation, and the consequences for development of atherosclerosis. The objectives are as follows: Aim #1. Determine the effects of iron ion and proton irradiation vs. gamma irradiation on expression of key adhesion molecules on cultured endothelial cells, both at baseline and following cytokine stimulation. Aim #2. Determine the consequences (for adhesion) of iron ion- and proton- irradiation-induced adhesion molecule expression and compare these to gamma irradiation using functional adhesion assays. Aim #3. Determine whether radiation accelerates atherosclerosis in vivo using the well characterized ApoE mouse model. Aim #4: Determine whether Tepoxalin reduces radiation-induced atherosclerosis. Key Findings: 1) Cultured human aortic endothelial cells were irradiated to determine the effects of radiation on endothelial cell adhesiveness, using both iron ions and protons. We found that both protons and iron ions cause small increases surface expression of specific adhesion molecules involved in monocyte-endothelial cell adhesion, as measured by flow cytometry. Impact: This represents major progress toward completion of Aim #1. We have determined that both iron ion and proton irradiation alter expression levels of key adhesion molecules in a way that is qualitatively consistent with promoting inflammation, yet with key differences from that typically produced by soluble inflammatory stimuli such as TNF-a. Subsequent experiments will further define differences and similarities between radiation-induced- and physiological inflammation. 2) A functional assay using a flow-chamber-based adhesion assay designed to mimic conditions in the bloodstream was used to measure adhesiveness of irradiated endothelial cells, following either iron ion or proton irradiation. In both cases, a clear effect of irradiation on adhesiveness was demonstrated. These adhesion increases were as pronounced as those produced by typical inflammatory stimuli, even though radiation-induced adhesion molecule expression were typically much smaller. Impact: This represents progress toward completion of Aim #2. The significance of this finding is that it clearly demonstrates that both iron and proton irradiation have the potential to increase vessel wall adhesivity. While the molecular mechanism is not yet clear, the end result (adhesion) is clearly pro-inflammatory. Therefore, the potential consequences for astronaut health on prolonged deep-space missions represent a risk that warrants further investigation. 3) ApoE -/- mice, a standard atherosclerosis model, were irradiated with 600 MeV 56Fe at 2 and 5 Gy in Spring 2007 and 2008 and with 200 MeV protons at 0.5 and 2 Gy in Fall 2007 and 2008. Mice were sacrificed at specific time points to determine whether iron ion and/or proton radiation accelerates development of atherosclerosis. Aortic plaques in irradiated mice were compared to age- and sex-matched un-irradiated controls were quantified. Iron-ion irradiation was found to accelerate development of atherosclerotic plaques. Results of proton experiments are pending. Impact: Demonstration of whether radiation accelerates development of atherosclerosis in this well-established mouse atherosclerosis model will have implications for whether astronauts on deep-space missions are at increased risk of developing vascular disease. Proposed Research for the coming year: 1. Flow-cell adhesion experiments at Brookhaven National Laboratory (BNL) to identify the key adhesion molecules involved in observed adhesiveness increases in response to 56Fe irradiation. This will be correlated with similar x-ray irradiation experiments (at UAB). 2. Continuing analysis of atherosclerotic plaque development in mice already irradiated. Experiments at BNL to refine radiation dose dependence of development of atherosclerosis. X-ray irradiation of mice (at UAB) for comparison. 3. When analysis of atherosclerotic plaques from previous experiments is complete, experiments to begin to test pharmacologic or dietary countermeasures will begin.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Among the non-cancer risks of radiation exposure are inflammatory diseases in general and atherosclerosis in particular. Increased risk of atherosclerosis is a known side effect of radiation exposure (generally gamma rays) on Earth, and increased endothelial adhesivity (an inflammatory process) is an important component of the mechanism. In this project, we determine whether iron ions and/or proton radiation pose vascular disease risks and begin to understand the mechanism and dose response. With therapeutic proton irradiation becoming more prevalent, the findings concerning protons in particular will have implications for cancer therapy. Since all results will be compared to similar experiments using gamma- or x-rays, a greater understanding of the risks posed by terrestrial radiation should also result. Additionally, we will determine the feasibility of using in vitro adhesion assays measuring acute changes in endothelial cell adhesivity to predict the risk of longer-term vascular damage. Finally, we will determine the feasibility of selected countermeasures. Thus, knowledge and insights gained in this project may also be used to prevent, counter or ameliorate biological damage from terrestrial radiation sources, such as nuclear weapons or radiation bioterrorism, or adverse

	effects from therapeutic radiation.
Task Progress:	1) Preliminary data last year indicated that both protons and iron ions can increase surface expression of specific adhesion molecules involved in monocyte-endothelial cell adhesion, as measured by flow cytometry. These experiments have now been repeated, with similar results. At this point, we have not determined whether adhesion molecule expression changes change adhesiveness, however. Adhesion assays will help to answer this question. 2) A flow-chamber based adhesion assay designed to mimic conditions in the bloodstream was used to measure adhesiveness of irradiated endothelial cells, following either iron ion or proton irradiation. We found that adhesiveness changes were as pronounced as those produced by typical inflammatory stimuli, and would be expected to have health consequences. In order to better understand the mechanism of the increased adhesion, we have begun experiments to identify the adhesion molecules and ligands responsible. We find that blocking specific adhesion molecules on leukocytes can modulate the pro-adhesive effect of radiation of endothelial cells, even if the radiation had only minor effects on endothelial-cell ligand surface expression. The significance of this finding is that it clearly demonstrates that both iron and proton irradiation have the potential to increase vessel wall adhesivity, which is an early step in development of inflammation. Therefore, the potential consequences for astronaut health on prolonged deep-space missions represent a risk that warrants further investigation. 3) ApoE -/- mice were irradiated using collimated beams focused on the aortic arches and carotids. Mice were given 600 MeV 56Fe at 2 and 5 Gy in Spring 2007 and 2008 and 200 MeV protons at 0.5 and 2 Gy in Fall 2007 and 2008. Mice were then sacrificed at 13 and 40 weeks post-irradiation to determine whether iron ion and/or proton radiation accelerated development of atherosclerosis. This year, in addition to repeating the radiation experiments, much of the analysis of last year's mice was completed. Other mice that have been recently irradiated have not yet reached the time point for analysis. 4) Radiation can have substantial negative effects on lymphoid organs, which could complicate interpretation of our atherosclerosis results. It is known that whole-body 56Fe irradiation of mice has long-lasting effects on circulating lymphocyte numbers. Whether these changes are due to radiation of the thymus, radiation of the bone marrow, or both cannot be determined from whole-body irradiation experiments. This question can be addressed by our experiments, however. To determine whether leukocyte populations would be affected by radiation to the chest and neck, we measured relative blood leukocyte populations at 13 and 40 weeks following irradiation with 56Fe at 2 and 5 Gy. We have now determined which leukocyte populations are affected by this targeted irradiation at each of these time points.
Bibliography Type:	Description: (Last Updated: 04/12/2018)
Articles in Peer-reviewed Journals	Khaled S, Gupta KB, Wu X, Yu T, Chang P, Kucik DF. "Iron ion irradiation alters adhesiveness of vascular endothelium." Gravitational and Space Biology. 2008 Jun;21(2):25-6. , Jun-2008